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WEATHER SATELLITE STATION
For Meteor M2 Reception

PRASANNA WAICHAL

The basic setup needed to receive the weather satellites was covered in 'Set Up Your Own Weather Satellite Receiving Station' article published in January issue. The antenna setup, the receivers, and decoding software were discussed in detail in that article. By this time, the author hopes that many readers would have their receiving stations up and running.

In this article, we shall look at the imaging systems of these satellites and how the automatic picture transmission (APT) protocol works. Later we shall see how to decode the digital QPSK transmission from the Russian Meteor M2 satellite and produce even higher resolution earth imagery.

It should be noted that the APT transmission is being phased out and the currently operational satellites, such as NOAA-15, NOAA-18, and NOAA-19, are the last ones to carry such analogue transmission payload. But why discuss this phased out system when digital transmission, such as Meteor, is readily available?

Many universities and academic institutions all over the world are building nano satellites (Cube Sats). Therefore, one can think of how meaningful and scientific imaging data can still be transmitted using such narrow-band protocol. May be some groups active with the nano satellites can work towards such an experiment in the future Cube Sat missions!

The present scope of discussion is:

- Basic imaging process onboard NOAA satellites
- The high-resolution picture transmission (HRPT) and low-resolution analogue APT transmission protocol
- Receiving and decoding weather images from Russian Meteor M2 satellite

TABLE 1
SPECTRAL BANDS AND THEIR APPLICATIONS

Channel No.	Wavelength range	Spectral band	Applications
1	580nm – 680nm	Visible light	Day-time reflected sunlight from earth's surface
2	725nm – 1000nm	Near IR	Reflected infrared energy from earth, used for contrasting land/ocean
3A	1580nm – 1640nm	Near IR	Contrast for snow/ice
3B	3.55µm – 3.93µm	Thermal	Reflected as well as thermal infrared generated from forest fire
4	10.20µm – 11.30µm	Thermal	Sea-surface temperature, night-time cloud mapping
5	11.5µm – 12.5µm	Thermal	SST and night-time cloud mapping

Source: Users Guide Building Receiving Station – NOAA, 2009

Basic imaging process

The images that we receive from these earth imaging satellites are generally a result of what is called scanning radio meter, an instrument that produces the output to the corresponding intensity of the spectral components through scanning. It utilises a mirror that rotates at angular speed of 360rpm and reflects the energy received from various lenses to the spectral sensors.

As the satellite progresses through its orbital path, the reflected energy received perpendicular to the satellite's trajectory is used to construct each line of the image. Thus, a complete two-dimensional image of the area over which the satellite passes is constructed.

As the mirror rotates during each scan, the sensors look at the earth using the mirror and lens arrangement. These sensors are calibrated on-the-fly during each line with respect to emission from the Deep Sky and Black-Body radiation on board the spacecraft as a reference. This entire instrumentation is called very high resolution radio meter (VHRR).

Table 1 summarises the various spectral bands and their primary applications.

As the energy/emission falls on these sensors, electrical signal is produced, which is further digitised by an analogue-to-digital converter (ADC) with a resolution of 10 bits, thereby producing an equivalent of 1024 grey-scale levels.

The transmission protocols

HRPT transmission. With a high-resolution picture transmission (HRPT) protocol, this image data is transmitted at 360 lines per minute and each of the scan line produces 2048 pixels per line. Since this is a very huge data, the signal is trans-